

Lavesh Babooram

Junior Network Engineer at Mauritius Telecom | B. Eng. (Hons) Telecommunications Engineering with Networking | MSc by Applied Research | Author

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- 📅 October 1998

Awards

1. Professor Soodursun Jugessur Gold Medal

University of Mauritius
January 2022

Gold Medal for best B. Eng. Final Year Project at the Faculty of Engineering 2020/2021.

2. Best Student Poster Award

IEEE Electrical Design of Advanced Packaging and Systems (EDAPS)
December 2023

For the paper: "A Web-based Network Traffic Monitoring and Analysis Application for Prediction and Classification using Machine Learning".

3. "My Bright Idea" Competition

Mauritius Telecom
May 2024

1st Prize for proposing and implementing a project titled "Smart Service Monitoring with Artificial Intelligence".

4. Gold Award (International)

49th International Convention on Quality Control Circles (ICQCC) 2024 Sri Lanka
November 2024

Presented a project implemented at Mauritius Telecom on the use of AI for smart service monitoring in telecommunications.

About me

I am a First Class graduate in Telecommunications Engineering with Networking (with Honours) of the University of Mauritius. I have been awarded the Gold medal for having produced the best B. Eng. Final Year project at the Faculty of Engineering for the academic year 2020/2021. I also hold an MSc by Applied Research in Telecommunications Engineering from the University of Mauritius, where my research focused on artificial intelligence and machine learning for network traffic analysis, QoS/QoE analytics, and real-time multimedia communications. With strong foundations in network infrastructure, service design, and data-driven analytics, my work lies at the intersection of telecommunications systems and artificial intelligence, with an emphasis on applied, industry-relevant research. My research output includes numerous technical books with journal and conference papers. Concurrently, I am working as a Junior Network Engineer at the Customer Experience and Service Department of Mauritius Telecom Ltd.

Education

University of Mauritius
MSc by Applied Research
July 2022 - Oct 2024

An AI Based Ubiquitous Network Traffic and QoS Analytics System for Real-Time Multimedia Communications

University of Mauritius
B.Eng. (Hons) Telecommunications Engineering with Networking
First Class Honours; CPA: 76.9
August 2017 - August 2021

Awarded the Gold Medal for producing the best B. Eng. Final Year Project at the Faculty of Engineering for the project:

Implementation and Performance Analysis of a Cloud Based Network Data Capture and Analytics Application

Royal College Curepipe
Higher School Certificate
November 2016

- Mathematics: A*
- Physics: A*
- Chemistry: A
- General Paper: a
- French: a

Royal College Curepipe
SC Certificate
November 2014

- Mathematics: 1
- Computer Studies: 1
- Chemistry: 1
- English Language: 1
- French: 1
- Physics: 2
- Additional Mathematics: 2
- Biology: 2

Skills

Telecommunications Application Development

Wireless Technologies

Artificial Intelligence

Machine Learning

Web Automation

Neural Networks

Deep Learning

Network Traffic Monitoring

Video Quality Analysis

LLMs and VLMs

Retrieval-Augmental Generation

Multimodal Model Integration

LangChain

IoT Solutions

Network Security

Communications Security

Optical Systems

Mobile Networks

Jira, Confluence, Docker, Git

Project Management

Android Development

Full Stack Development

Languages

English

5/5

French

5/5

Mauritian Creole

5/5

Hindi

4/5

Experience

Mauritius Telecom Ltd., Port Louis

October 2022 - Present

Junior Network Engineer

- Oversee end-to-end telecommunications services for enterprise customers in Mauritius and international markets.
- Perform service management, implement monitoring tools, detect faults, and prevent SLA breaches.
- Drive initiatives to enhance the monitoring and management of Telecommunications services, such as voice, mobile, broadband internet, and IPTV.
- Conduct research and propose new services that align with industry trends while catering for evolving customer demands.
- Develop service monitoring and quality assurance tools for over-the-top (OTT) platforms.
- Collaborate with marketing and technical teams to evaluate and recommend new devices for commercialisation, ensuring alignment with customer needs and technological advancements.
- Design customer-centric solutions for Wi-Fi setups, optimising home and business network environments based on unique requirements.

University of Mauritius

March 2024

Short-Course Trainer

- Design and deliver training modules covering foundational concepts of AI, machine learning, and web development to participants with varying levels of expertise.
- Facilitate interactive sessions to teach HTML, JavaScript, Node.js and data preprocessing techniques.

Infosys Ltd., Ebene

October 2021 - July 2022

Associate

- Training in advanced Java and SQL for web project development.
- Handle multiple projects concurrently for an automotive company.
- Maintain, debug and correct application codes within established protocols, guidelines and quality standards to meet client demands.
- Develop technical documentation to streamline maintenance and onboarding processes for future developers.

Mauritius Telecom Ltd., Port Louis

July 2020 - August 2020

Intern

- Complaints management, fault debugging and software-oriented overview of networking architecture.
- Creation of electronic records with data from on-site planners.
- Exposure to Fiber to the Home (FTTH) architecture.
- Exposure to Optical Transmission Networks and Optical Distribution Networks.
- Inspection, planning, assistance and survey during provisioning.

Projects

Interactive Cloud-based Machine Learning Website
2020

- https://lav-machine-learning.web.a...
- Automating browser-related systems for manipulating statistical data using JavaScript and Firebase.

Browser Extensions for WebRTC and Security Analysis
2020

- Web scraping to detect spamming intentions and abnormal activities.

AI-Powered Internet Service Monitoring (at Mauritius Telecom)
(International Award Winning Project)
2024-2025

- Designed and deployed a real-time quality of experience framework for live video monitoring across Mauritius.
- Developed automation scripts to simulate end-user interactions and monitor video availability, and quality scores, with ML to predict outages and degradations.

Volunteering

Institute of Electrical and Electronics Engineers
Research Paper Reviewer
December 2024

- Peer-reviewed the following papers:
- 5G-Advanced infrastructure for Smart Energy: Converging technologies and management
 - Harnessing Named Entity Recognition to Extract Key Information from Unstructured Documents
 - Evidential Reasoning Approach for Predicting Popularity of Instagram Posts

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Publications

Book: AI Solutions for the United Nations Sustainable Development Goals (UN SDGs)
Apress (Springer)

Theoretical knowledge and hands-on experience in comprehending and constructing machine learning-based applications for addressing multiple UN SDGs using JavaScript. [\[More\]](#)

Book: Machine Learning For Network Traffic and Video Quality Analysis
Apress (Springer)

Theoretical insights and hands-on experience in understanding and building machine learning-based Network Traffic Monitoring and Analysis (NTMA) and Video Quality Assessment (VQA) applications using JavaScript. [\[More\]](#)

Paper: Performance analysis of collaborative real-time video quality of service prediction with machine learning algorithms
Springer's International Journal of Data Science and Analytics
[\[More\]](#)

Book Chapter: Leveraging the Power of Blockchain in Industry 4.0 and Intelligent Real-Time Systems for Achieving the SDGs
Emerald Publishing Limited
[\[More\]](#)

Paper: Implementation and Performance Analysis of a SIP Enabled Softphone System with Machine Learning Based QoS Assessment
2023 Global Conference on Information Technologies and Communications (GCITC)
[\[More\]](#)

Book Chapter: Driving 5G Mobile Communications with Artificial Intelligence towards 6G
Routledge and CRC Press (Taylor & Francis Group)
Artificial Intelligence towards 6G - Enabling technologies and applications of 5G/6G powered intelligent connectivity. [\[More\]](#)

Paper: A Web-Based Network Traffic Prediction and Classification Application using Machine Learning
Institute of Electrical and Electronics Engineers (IEEE)
2022 4th International Conference on Emerging Trends in Electrical, Electronic and Communications Engineering (ELECOM). [\[More\]](#)

Paper: Performance analysis of a cloud-based network analytics system with multiple-source data aggregation
Emerald Publishing
Network Traffic Monitoring and Analysis using Machine Learning and Deep Learning techniques, and software coded in Java on Android and PC. [\[More\]](#)

Book: Real-Time Cloud Computing and Machine Learning Applications
Nova Science Publishers
Developed and documented a series of applications running on both Android and Desktop to serve as networking monitoring and analytics tools, using an array of machine learning and deep learning algorithms for network behaviour prediction and classification. [\[More\]](#)